Experiment # 10

Introduction to Telnet and SSH

Objective:

Learn how to remotely access a router using Telnet and SSH techniques.

Software Tools:

- Cisco Packet Tracer

Theory:

We can take access to a cisco router or switch either through a console cable or taking remote access through well-known protocols Telnet or SSH (Secure Shell). Telnet and SSH are both application layer protocols used to take remote access and manage a device.

Telnet:

The Telnet technique is used to remotely access a router from a PC. We can remotely access any router in a network. The techniques involve opening of virtual terminals by which we identify how many PCs can remotely access at a time. Telnet is an application layer protocol which uses TCP port number 23, used to take remote access of a device.

Features

1. It doesn't support authentication.
2. Data is sent in clear text therefore less secure.
3. No encryption mechanism is used.
4. Designed to work in local networks only.

Configuring telnet:

```
Router1(config)#line vty 0 4

Router1(config-line)#password root

Router1(config-line)#exit
```

Here, 0 4 means that we can have 5 concurrent sessions at a time.

SSH:

This technique is also used to remotely access a router but involves encryption. It consists of high level of encryption. The level of encryption required can be set while setting ssh. The message goes in encrypted form so only the required person has the key to read it.

Features

1. Unlike telnet, it provides authentication methods.
2. The data sent is in encrypted form.
3. It is designed to work in public networks.
4. It uses public keys for encryption mechanism.

In short, SSH is more secure than telnet.

Configuring ssh on Router1.

```
Router(config)#ip domain-name GeeksforGeeks.com
Router(config)#hostname Router1
Router1(config)#line vty 0 4
Router1(config-line)#transport input ssh
Router1(config-line)password root
Router1(config-line)#login
Router1(config)#crypto key generate rsa label Root
modulus 1024
```

Here, note that it is necessary to give domain name as ssh uses it and password for encryption purpose. If domain-name and hostname are not provided then crypto keys will not be generated. We have provided a password for vty line login and at last we have created key of 1024 bytes and labelled it as Root.

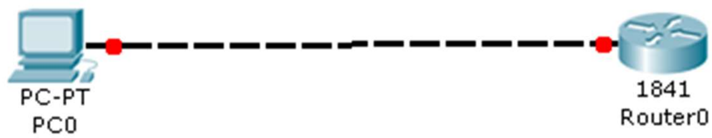
Another Method:

```
Router>en
Router#conf t
Router(config)#hostname R1
R1(config)#ip domain-name www.google.com
R1(config)#crypto key generate rsa
R1(config)#username root password root
R1(config)#line vty 0 4
R1(config-line)#transport input ssh
R1(config-line)#login local
R1(config-line)# exit
```

Username and password configured on router locally and not on vty lines then login local command will be used.

Example:

The tasks we performed in this lab are as follow:-



Steps:

1. Set pc0 IP address to 192.168.1.2/24
2. Set interface fast Ethernet 0/0 IP address to 192.168.1.1/24
3. Set privileged mode password to Cisco
4. Enable telnet lines on router
5. Test telnet connection via your PC

Configurations

```
Router(config)#interface fastEthernet 0/0

Router(config-if)#ip address 192.168.1.1 255.255.255.0

Router(config-if)#no shutdown

Router(config-if)#exit

Router(config)#enable password cisco

Router(config)#line vty 0 15

Router(config-line)#password cisco

Router(config-line)#login

Router(config-line)#exit
```

Click on pc0 > click on command prompt > type telnet 192.168.1.1 > type privileged mode password

Configuring SSH on a router in Packet Tracer

Configuring SSH on the router so that you as the admin can access and manage it remotely using an SSH client on the admin PC. First build the network topology.

Then do these basic IP configurations on the PC and the router:

Router

```
Router(config)#int fa0/0

Router(config-if)#ip add 10.0.0.1 255.0.0.0

Router(config-if)#no shut

Router(config)#exit
```

PC : **IP address** 10.0.0.10 **Subnet mask** 255.0.0.0 **Default gateway** 10.0.0.1

Now, to set up SSH on the router, you'll need to:

1. Set Router's **hostname**

```
Router(config)#hostname myRouter
```

2. Set **domain name**

```
myRouter(config)#ip domain name admin
```

Both the *hostname* and *domain name* will be used in the process of **generating encryption keys.**

3. Now generate **encryption keys** for securing the session using the command *crypto key generate rsa.*

```
myRouter(config)#crypto key generate rsa
```

```
The name for the keys will be: myRouter.admin
Choose the size of the key modulus in the range of 360 to 2048 for your

General Purpose Keys. Choosing a key modulus greater than 512 may take

a few minutes.

How many bits in the modulus [512]: 1024

% Generating 1024 bit RSA keys, keys will be non-exportable...[OK]
```

4. Set an **enable password .**

```
myRouter(config)# enable password admin
```

Note that this password is not for use with SSH; its only for use in accessing the privileged executive mode of the router after you are able to access its CLI remotely via SSH .

5.Set **username** and **password** for local login.

```
myRouter(config)#username admin password admin
```

The password will have to be provided before you can access the CLI of the router when using SSH.

6.Specify the **SSH version** to use.

```
myRouter(config)#ip ssh version 2
```

7.Now connect to **VTY** lines of the Router and configure the SSH protocol.

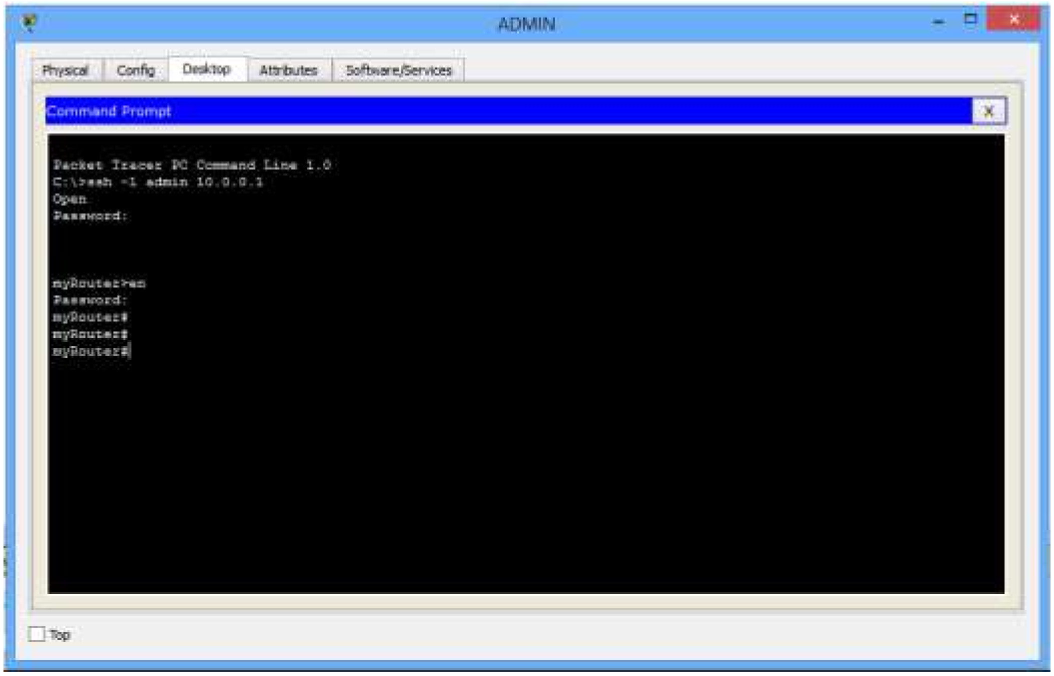
```
myRouter(config)#line vty 0  15

myRouter(config-line)#transport input ssh

myRouter(config-line)#login local
```

That's all for configuration. Move on to see if you can access the router remotely from the PC.

8. On the command prompt of the **PC**, open a SSH session to the remote router by typing the command: ***ssh -l admin 10.0.0.1***

admin is the **username** set in step 5.

9. Provide the **login password** which you set in step 5 and press enter. You're now probably in the CLI of the router. Provide the **enable password** (the one you set in step 4) to access the privileged executive mode.



You can proceed and do configurations on the Router. You're now managing the router remotely from the PC.

Next comes the configuration of SSH on a **switch**.

SSH configuration on a Switch

Here, we'll configure SSH on a multi-layer switch. The commands remain almost the same as for the router; only that in a switch, we'll use the IP address of its **VLAN interface** to access it from the PC. Begin by creating the following network topology.

Then configure basic IP addressing on the PC and the switch. On the switch, we'll assign an IP address to a VLAN interface, just as we've said.

Switch

```
Switch(config)#int vlan 1

Switch(config-if)#ip add 10.0.0.1  255.0.0.0

Switch(config-if)#no shut
```

Give the **ADMIN PC** IP address 10.0.0.10 /8

Now, to configure SSH on the multilayer switch, here are the steps.

1.Configure **hostname**

```
Switch(config)#hostname SW1
```

2. Configure IP **domain name**

```
SW1(config)#ip domain name admin
```

Both the host name and domain name will be used in the process of generating encryption keys.

3. Now generate **encryption keys** for securing the session.

```
SW1(config)#crypto key generate rsa

The name for the keys will be: SW1.admin

Choose the size of the key modulus in the range of 360 to 2048 for your

General Purpose Keys. Choosing a key modulus greater than 512 may take

a few minutes.

How many bits in the modulus [512]: 1024

% Generating 1024 bit RSA keys, keys will be non-exportable...[OK]
```

4.Set an **enable password.**

```
SW1(config)#enable password admin
```

Again, note that enable password is not necessarily used in configuring SSH; it will allow the admin to access the **privileged executive mode** of the switch once a remote connection to the switch via SSH is established.

5. Set **username** and **password** for local login.

```
SW1(config)#username admin password admin
```

6. Specify the **SSH version** to use.

```
SW1(config)#ip ssh version 2
```

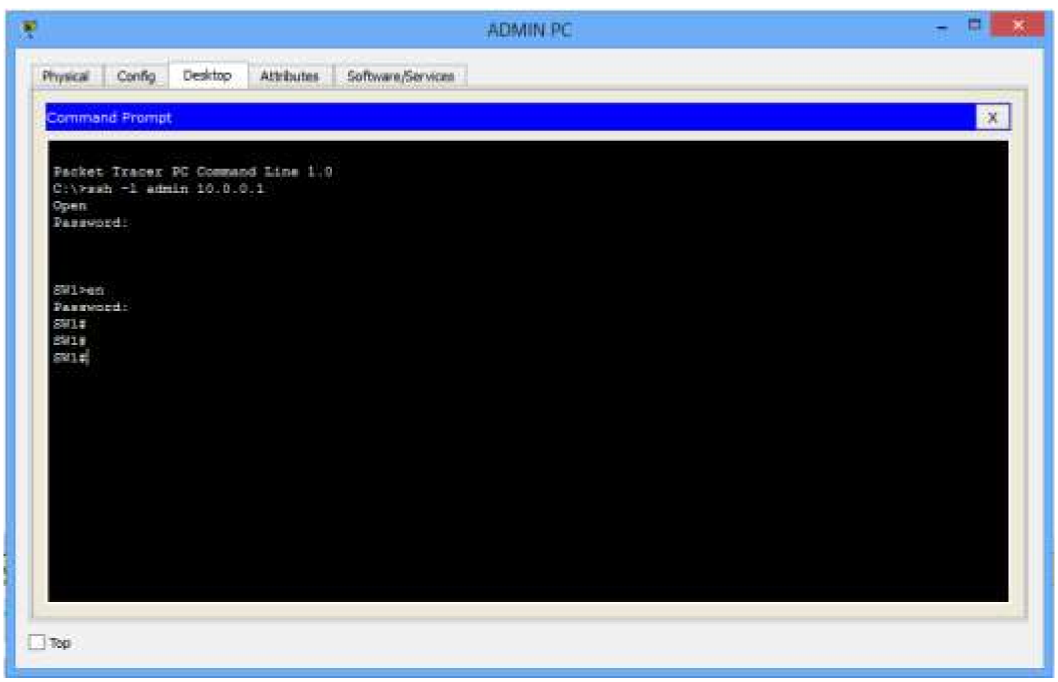
7. Now connect to the **VTY** lines of the switch and configure SSH on the lines.

```
SW1(config)#line vty 0 15

SW1(config-line)#transport input ssh

SW1(config-line)#login local
```

That’s all for SSH configuration on the switch. Move on and try to access the switch remotely from the PC.

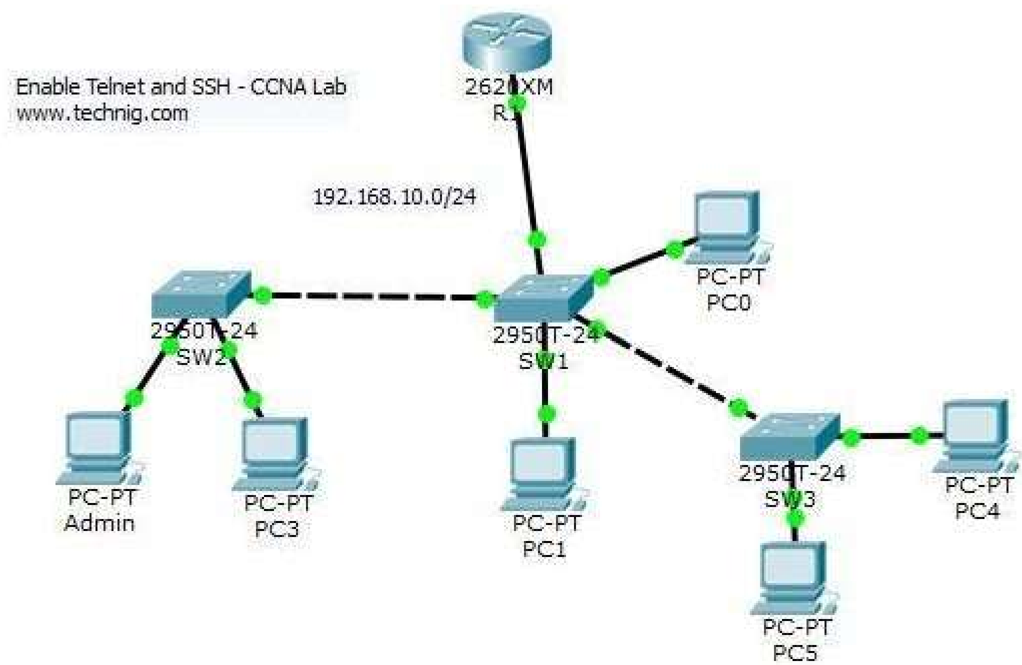
8. On the command prompt of the **Admin PC**, open a SSH session to the switch using the command **ssh -l admin 10.0.0.1**

Note that: ***admin*** is the username defined in step 5 while **10.0.0.1** is the IP address of the VLAN interface of then switch.



Lab Task

1. Create following topology and simulate the implementation of Telnet and SSH.



Conclusion

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